

Label-free Industrial Fault Detection via Adversarial Inverse Reinforcement Learning: A System for Run-to-Failure Prognostics

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Abstract

Machinery fault detection (MFD) remains heavily reliant on supervised learning, which struggles with the scarcity of fault labels in real-world settings. While reinforcement learning (RL) offers a framework to model the sequential nature of degradation, current “RL-based” MFD methods reduce the problem to a static contextual bandit (CB) formulation: by ignoring state transitions and discarding the temporal discount factor, they collapse to standard supervised classification. We propose an adversarial inverse reinforcement learning (AIRL) framework that treats MFD as an offline IRL problem. Unlike reconstruction-based approaches that rely on static error margins, or CBs that ignore dynamics, our method recovers an intrinsic “health” reward directly from observational state transitions, requiring neither manual reward engineering nor fault labels. On three run-to-failure benchmarks (HUMS2023, IMS, XJTU-SY), AIRL is the only method pairing within-window detection with non-saturated post-detection consistency across all datasets, while CB baselines fail to detect gradual degradation and reconstruction models collapse into always-anomalous states. Code and data: <https://github.com/dhirajneupane/AIRL-MFD-DN>.

CCS Concepts

• **Computing methodologies** → **Inverse reinforcement learning; Anomaly detection**; • **Applied computing** → **Industry and manufacturing**.

Keywords

Adversarial Learning; Anomaly Detection; Machinery Fault Detection; Reinforcement Learning; Machine Condition Monitoring

ACM Reference Format:

Dhiraj Neupane, Mohamed Reda Bouadjenek, Richard Dazeley, and Sunil Aryal. 2026. Label-free Industrial Fault Detection via Adversarial Inverse

Reinforcement Learning: A System for Run-to-Failure Prognostics. In *Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26)*, November 07–11, 2026, Rome, Italy. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3799682.3840145>

1 Introduction

Machinery fault detection (MFD) is essential for industrial reliability, yet acquiring labelled fault data for every fault type, severity, and component remains prohibitively expensive. Supervised learning dominates the field ($\approx 81\%$ of studies, vs. just 1% for RL [23]) but inherits this labelling bottleneck. Semi-supervised and unsupervised methods partially mitigate it [24, 27, 33, 52] but struggle with robustness under variable operating conditions. Reinforcement learning (RL) is a promising alternative through interactive, sequential decision-making, yet current applications neglect the temporal structure of fault progression [23].

Existing “RL-based” MFD methods [8, 16, 17, 32] reduce fault detection to a static *contextual-bandit* (CB) trial: each sensor segment is treated as an independent state, the agent issues a one-shot label, the discount factor γ is effectively zero, and episodes terminate after a single action. The sequential premise of RL therefore collapses into supervised classification. Moving beyond this requires multi-step interactions ($\gamma > 0$) and rewards reflecting long-term outcomes, but designing such rewards is difficult: fault progression is subtle, system dynamics are only partially understood, and most industrial datasets are prerecorded, making online RL infeasible. Even existing offline RL approaches still rely on manual reward design.

These challenges highlight the potential of inverse reinforcement learning (IRL) [26], which bypasses manual reward specification by learning directly from healthy operational trajectories offline. Despite its ability to capture normal machinery behaviour, IRL remains unexplored in machinery fault detection (MFD). To address this gap, we introduce adversarial inverse reinforcement learning (AIRL) [9] for MFD, marking its first application in this domain. AIRL operates fully offline, learning a reward model of normal sequential patterns to detect anomalies as deviations from these dynamics. This approach retains RL’s strengths while providing a robust, practical solution. Extensive experiments across multiple run-to-failure datasets demonstrate that our framework achieves earlier fault detection and more stable anomaly scoring than conventional one-class and reconstruction-based methods.



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CIKM '26, Rome, Italy

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ACM ISBN 979-8-4007-2539-5/2026/11

<https://doi.org/10.1145/3799682.3840145>

Applied contributions. A two-page extended abstract of this formulation appeared at AAMAS 2026 [25] with preliminary HUMS2023 results. That version reported detection timing for HUMS2023 only against seven baselines, with IMS and XJTU-SY noted as consistent trends. This paper adds full results on all three datasets, four further baselines (TCAE, CTCAE, USAD, Anomaly Transformer), a pre-onset false-alarm analysis with an explicit saturation-failure criterion, a comparison of ten thresholding rules of seven strategy families, a cost analysis, and the discount-factor ablation. Our specific contributions are: (1) a label-free detection pipeline applying the state-only imitation learning (SOIL) approach [40] to offline industrial sensor sequences; (2) validated generalization across three distinct real-world run-to-failure datasets (HUMS2023, IMS, XJTU-SY); (3) an extended baseline comparison that demonstrates why AUROC is structurally inapplicable, alongside a novel post-detection consistency (PDC) analysis of *saturation failure*; (4) an evaluation of seven automatic thresholding strategies with practical deployment recommendations; (5) a computational cost analysis validating AIRL’s compact architecture for edge deployment; and (6) an ablation study isolating the role of the discount factor in sequential modeling.

2 Background and Related Works

2.1 Machinery Fault Detection Problem

Industrial machinery often operates under harsh conditions, resulting in gradual faults, such as wear, cracks, and broken teeth, that vary in type, severity, and the components they affect [28]. As shown by Neupane et al. [24], most publicly available datasets reflect this variety, yet are strictly offline and designed for classification tasks with predefined labels. However, in real-world settings, healthy data is abundant, whereas acquiring and labeling faulty data for every possible failure mode is a significant challenge. Consequently, treating MFD as an anomaly detection task, training on abundant normal data to flag deviations at test time, is a convincing and practical approach. While the current research trend remains heavily skewed towards fault classification under supervised learning, robust fault detection remains the foundational necessity in industrial applications.

2.2 Adversarial Inverse Reinforcement Learning

Reinforcement learning models sequential decision-making via Markov decision processes (MDPs), optimizing policies to maximize cumulative reward [39]. However, designing reward functions that precisely encode desired behaviors is challenging, particularly for complex tasks like MFD. IRL addresses this by inferring the underlying reward directly from expert demonstrations [7]. While traditional IRL struggles with high-dimensional environments, AIRL [9] scales effectively by employing a generative adversarial framework. AIRL trains a discriminator to differentiate between expert (healthy) and policy-generated trajectories, simultaneously updating a policy to imitate the expert behavior. Importantly, AIRL learns a robust reward function that is disentangled from the environment’s specific dynamics, making it highly suitable for anomaly detection where explicit reward formulation is impractical. The discriminator’s output is formulated as [9]:

$$D(s, a, s') = \frac{\exp(r(s, a) + \gamma V(s') - V(s))}{\exp(r(s, a) + \gamma V(s') - V(s)) + \pi(a | s)}, \quad (1)$$

where (s, a, s') is the transition tuple consisting of the current state s , action a , and next state s' , $r(s, a)$ is the learned reward, $V(s)$ the state-value function, $\pi(a | s)$ the stochastic policy, and γ the discount factor. This architecture ensures the recovered reward remains invariant to dynamic shifts, providing a stable basis for evaluating operational health.

2.3 RL vs. Contextual-Bandit Practice in MFD

The application of RL to rotating machinery fault detection has gained popularity since 2019, as summarized in Table 1. However, a consistent pattern emerges: despite the variety of agents and reported episode lengths, these studies formulate the problem as a single-step classification task rather than a true MDP. Specifically:

- Datasets are typically pre-shuffled, treating each sensor signal segment as an *iid* (independent and identically distributed) context. The agent predicts a single label, and the episode immediately terminates.
- Because transitions (s, a, s') are ignored, the discount factor γ (even if nonzero) plays no mathematical role in credit assignment.

Formally, these setups are *single-step contextual bandits* (CBs). Even when studies report episode lengths of several hundred steps (Table 1), each step remains an independent sample with no dependency on previous actions. The Bellman equation collapses into a supervised classifier optimized via a reward-based loss. While this approach is valuable for static labelling, using RL terminology misrepresents the methodological novelty. More importantly, it fails to exploit RL’s core strength: modelling sequential system evolution for fault *progression* and *early-warning* objectives.

3 AIRL Framework for MFD

Building on the problem formulation and related work discussed earlier, this section presents the proposed adversarial inverse reinforcement learning framework for machinery fault detection. The framework uses preprocessed vibration sequences to autonomously learn reward dynamics that characterize normal operational behavior and deviations from it. It consists of three main components: (i) transition construction from vibration signals, (ii) neural architectures for reward, value, and policy estimation, and (iii) adversarial training and threshold-based anomaly scoring.

3.1 Transition Construction for AIRL

Let $X = \{x_1, x_2, \dots, x_L\}$ denote a vibration record of length L samples, and let $\tilde{x}_t$ denote the z -normalized version of x_t using statistics computed solely from the healthy training partition (to prevent test-train leakage). Records are sorted chronologically and split into a healthy expert set $\mathcal{D}_{\text{exp}}$ used for training and a held-out test set $\mathcal{D}_{\text{test}}$ containing the degradation phase. Dataset-specific partitions, record lengths, and acquisition details are reported in Section 4.

To construct the AIRL input, state-action-next-state tuples are generated from each normalized sequence:

$$s_t = \tilde{x}_t, \quad a_t = \tilde{x}_{t+1}, \quad s_{t+1} = \tilde{x}_{t+1}. \quad (2)$$

Table 1: RL formulations in recent MFD research.

Ref/Year	State	Action	γ	Ep. Len.	Agent
[8]/19	Random	Label	0	1	SAE
[14]/21	Random	Label	-	1	CapNet
[32]/22	Random	Label	0	1	CNN-GRU
[49]/23	Random	Label	0.1*	500 [†]	ConvResAE
[47]/24	Random	Label	0.1*	504 [†]	1D-CNN
[17]/25	Random	Label	0.99*	1	Eq. DQN
[16]/25	Random	Label	0.99*	1	ConvTrans
[36]/25	Random	Label	0.1*	1	Duel-DQN

Note: * γ not used for credit assignment. [†] Samples pre-shuffled.

Since the data is offline and purely observational, no recorded control actions are available. We adopt the SOIL approach [40] and treat the next observation $\tilde{x}_{t+1}$ as a proxy action, effectively learning an inverse-dynamics model that scores the plausibility of state transitions (s_t, s_{t+1}) rather than maximizing a static classification reward. This aligns with state-only imitation and inverse-RL methods such as GAIfo [41], Diffusion-IfO [10], and OPOLO [54], which optimize a discriminator over state transitions rather than explicit control actions. The construction enables AIRL to model the temporal evolution of healthy and faulty sequences without introducing artificial control variables, i.e., nothing is being controlled, and the word “action” is only the label attached to the observed next value so that a standard AIRL discriminator applies unchanged.

3.2 Model Architecture

The proposed AIRL framework comprises three neural components: a reward network $r(s, a)$, a value network $V(s)$, and a stochastic policy $\pi(a|s)$. Note that while we denote the input as a to maintain standard RL notation, a corresponds to the next state $\tilde{x}_{t+1}$. Each network is a two-layer MLP of the form $f(x) = W_2 \phi(W_1 x + b_1) + b_2$, where $\phi(\cdot)$ is the ReLU activation, the hidden layer has 64 units, and b_1, b_2 are learnable biases. The networks differ only in their input dimension and output role:

- **Reward network (r):** Maps the concatenated state-action vector $[s; a] \in \mathbb{R}^2$ to a scalar reward distinguishing healthy (expert) from faulty (policy-generated) transitions.
- **Value network (V):** Takes scalar state $s \in \mathbb{R}$ and outputs the expected cumulative reward, providing a potential-based shaping term for reward generalization.
- **Policy network (π):** Takes scalar state $s \in \mathbb{R}$ and outputs the mean $\mu(s)$ of a Gaussian policy $\pi(a|s) = \mathcal{N}(\mu(s), \sigma^2)$; the state-independent log-standard-deviation $\log \sigma$ is learned separately. This stochasticity encourages exploration during roll-outs and mitigates overfitting to logged demonstrations.

The total parameter count is 644 (reward: 257, value: 193, policy: 194).

3.3 Training Procedure

Each AIRL training epoch consists of two phases: (i) discriminator learning and (ii) policy improvement. During discriminator learning, mini-batches of 128 expert and policy-generated transitions are

drawn, and the reward and value networks are jointly updated by minimizing the standard AIRL cross-entropy loss:

$$\mathcal{L}_{\text{disc}} = - \left[\log \sigma(r_\phi(s, a) + \gamma V_\psi(s') - V_\psi(s)) + \log \left(1 - \sigma(r_\phi(s, a_\pi) + \gamma V_\psi(s'_\pi) - V_\psi(s_\pi)) \right) \right] \quad (3)$$

where $\sigma(\cdot)$ is the logistic function and $\gamma = 0.99$.

During policy improvement, π is rolled out in the learned reward landscape to estimate the REINFORCE gradient:

$$\nabla \mathbb{E}_\pi[r(s, a)] = \mathbb{E}_\pi[\nabla \log \pi(a|s) G_t], \quad (4)$$

using 50-step returns G_t , alongside an entropy bonus $\beta = 0.01$ to encourage exploration. All networks are optimized with Adam ($\alpha = 10^{-3}$) for $N_{\text{epochs}} = 300$ with 50 policy episodes per epoch.

3.4 Anomaly Scoring and Thresholding

Within AIRL, the discriminator assigns each transition (s, a, s') a confidence $D_\phi(s, a, s')$, which we convert to an anomaly score:

$$\alpha(s, a, s') = 1 - D_\phi(s, a, s'), \quad (5)$$

where higher values indicate greater deviation from expert behavior. Let $\{\alpha_i\}_{i=1}^N$ denote scores computed on *healthy* training transitions. We construct data-driven thresholds τ from these healthy scores and flag a test sample as anomalous when $\alpha > \tau$ (upper-tail rule).

Seven strategy families are evaluated: (1) mean+ $k\sigma$ ($k \in \{2, 3\}$) [22]; (2) median+3MAD [34]; (3) max-based variants (max, max-SE, max-2SE) [50]; (4) percentile p_{99} [42]; (5) Otsu’s method [29]; (6) k -means midpoint [21]; (7) GPD tail fit (top 5% of $\{\alpha_i\}$) [31]. All thresholds are computed once from healthy training scores and fixed at test time.

4 Experimental Section

4.1 Datasets and Preprocessing

We evaluate the proposed framework on three real-world run-to-failure datasets encompassing distinct machinery types, operating conditions, and fault modes. All methods use healthy-only training partitions and mixed (healthy + faulty) test partitions, as detailed in Table 2.

HUMS2023 [45]: This dataset captures the progressive degradation of a full-scale helicopter gearbox under authentic load conditions. Fatigue cracks were initiated via EDM notches and propagated under 125% torque. Hunting Tooth Synchronous Averaging (H-SSA) was utilized to align the vibration patterns with gear-meshing cycles [37]. The ground truth degradation onset is verified by the challenge committee as Day 24 (sample #264).

IMS Test 1 [35]: This is a widely used bearing dataset recorded at 20 kHz across 8 channels, from which Channel 1 is utilized for our evaluation. The ground truth degradation onset is established in the literature as mid-November [4, 35], which eventually culminates in catastrophic failure on Nov 25.

XJTU-SY [44]: This dataset features accelerated bearing degradation under heavy operational loads (Bearing 3_1; 40 kHz, 10 kN). The ground truth degradation onset is formally defined at ≈ 500 min [20].

Table 2: Training and testing splits.

Dataset	Training (healthy)	Testing (mixed)
HUMS	Day 17–20; Total: 282	Day 21–27; Total: 526
IMS	Oct 1–31; Total: 1,840	Nov 1–25; Total: 8,940
XJTU-SY	First 2,000 (~250 min)	Remaining 18,295

To maintain consistent temporal resolution and input dimensionality across the entire framework, all raw vibration sequences are z-normalized and segmented into uniform observation windows (4095 points for HUMS H-SSA samples, and 4096-point segments for IMS and XJTU-SY, yielding 5 segments per IMS sample and 8 per XJTU-SY sample).

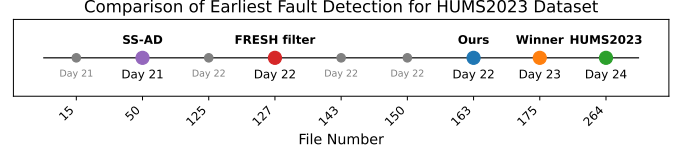
4.2 Baseline Methods

For comparison with AIRL, we trained a set of widely used unsupervised and one-class methods using only healthy data and evaluated them on both healthy and faulty segments. These baselines encompass a range of approaches, from traditional machine learning to deep learning models designed to capture temporal dependencies.

4.2.1 Unsupervised and One-Class Baselines. All baselines are trained on healthy data only and scored with the same thresholding rules as AIRL.

- **IF** (Isolation Forest) [18] and **OCSVM** (One-Class Support Vector Machine) [38]: classical one-class methods. IF: 400 estimators, max_samples=256; OCSVM: RBF kernel, $\nu = 0.05$.
- **AE** (Autoencoder) [53] and **VAE** (Variational Autoencoder) [12]: reconstruction-error anomaly scorers (hidden dims {1024,256,64}, latent 64). (~8.95M / ~8.97M params)
- **LSTM-AE** and **LSTM-VAE** [6, 19]: Long Short-Term Memory (LSTM) recurrent variants of AE/VAE for sequence modeling. (LSTM hidden=128, latent=64; ~167K / ~184K params)
- **TCAE** (Temporal Convolutional Autoencoder) [2]: 1D convolutional AE with strided convolutions (spatial 4096 $\rightarrow$ 64 via three stride-4 Conv1d layers, kernel 9, channels 1 $\rightarrow$ 16 $\rightarrow$ 32 $\rightarrow$ 8). (~14K params)
- **CTCAE** (Contrastive TCAE) [5]: TCAE + SimCLR contrastive loss on noise-augmented views. (wider bottleneck: channels 1 $\rightarrow$ 16 $\rightarrow$ 32 $\rightarrow$ 64, ~47K backbone params)
- **USAD** (Unsupervised Anomaly Detection) [1]: dual-decoder adversarial AE; anomaly score combines both decoders' outputs (~13.4M params).
- **ATF** (Anomaly Transformer) [48]: uses association-discrepancy between a Gaussian prior and learned softmax attention; ConvStem tokenizes inputs into 91 tokens for HUMS (4095 pts) and 92 tokens for IMS/XJTU-SY (4096 pts) (2 blocks, 64 dim, 4 heads, ~146K params).

4.2.2 Contextual-Bandit Baseline. We reproduce the LiteDPER-CTQN agent [16] (reimplemented in PyTorch) as a representative of the common RL-as-classifier pattern in MFD. It trains on balanced Normal/Faulty splits (HUMS: Days 19–20 vs. 26–27; IMS: 1370 Oct segments vs. last 1370 Nov segments; XJTU-SY: first 2000 vs. last 2000 segments) and is tested on the remaining unlabeled files.

**Figure 1: Comparison of Earliest Fault Detection for HUMS2023 Dataset**

4.3 Evaluation Criteria

Our primary objective is evaluating *earliest detection*, i.e., how soon a model reliably signals anomalies within the incipient degradation window. Because early warnings are operationally useless alongside continuous false positives, we also assess the pre-onset *False Alarm Rate* (FAR) and *post-detection consistency* to ensure the stability of valid alerts.

Why AUROC is inapplicable: Run-to-failure data lacks per-sample binary labels, and retrospective fault onsets are inherently uncertain (e.g., the HUMS2023 onset spans Days 22–24). Consequently, AUROC computed against a rigid boundary is fragile and indifferent to detection earliness. Following standard practices [13, 15], we instead evaluate reliability via *timeliness*, *FAR*, and *PDC*, explicitly flagging *saturation failure* (achieving high PDC merely through persistent, indiscriminate alerting).

4.4 Experimental Setup

All models use the splits in Table 2, trained over ten random seeds (scores averaged). AIRL uses the hyperparameters in Section 3.3. AE-family baselines use symmetric encoder-decoders with hidden dims {1024,256,64}, latent 64. IF uses 400 estimators; OCSVM uses an RBF kernel. All methods share the same thresholding rules (Section 3.4).

5 Evaluation

5.1 Earliest Fault Detection

Table 3 summarizes the earliest anomaly detections across the three datasets. On HUMS, nearly every baseline triggered on Day 21, well before the committee-verified onset of Day 24 #264, indicating premature false alarms. CTQN exhibited mode collapse: all 416 HUMS test files (Days 21–25) flagged faulty with a near-constant score ≈ 0.546 , while CTCAE failed in the opposite direction, producing no detections at all. AIRL detected at Day 22 #163, late enough to clear the false-alarm zone, early enough to precede committee ground truth. Figure 1 illustrates this against prior HUMS2023 work: AIRL falls between FRESH filter [43] (Day 22 #127) and the official challenge winner [30] (Day 23 #175), aligning with the committee-verified onset.

On IMS, evaluated against the mid-November onset [4, 35] (catastrophic failure Nov 25), baselines split into three failure modes. OCSVM, AE, USAD, and CTQN triggered Nov 1, with TCAE on Nov 3, indicating pre-GT false alarms days before any physical degradation. IF and ATF (Nov 8 and Nov 14) fall earlier than the established window. Conversely, VAE, LSTM-AE, and LSTM-VAE first flag only on Nov 22, missing the prognostic window entirely. AIRL alone triggered on Nov 17, precisely within the mid-November

Table 3: Earliest fault detection across datasets (max-2SE threshold).

Model	HUMS	IMS	XJTU-SY
GT	Day 24 #264	mid-Nov	~500 min
IF	Day 21 #9	2003-11-08 11:41:44	279
OCSVM	Day 21 #1	2003-11-01 00:41:44	264
AE	Day 21 #1	2003-11-01 05:31:44	309
VAE	Day 21 #37	2003-11-22 17:36:56 [†]	332
LSTM-AE	Day 21 #36	2003-11-22 17:36:56 [†]	332
LSTM-VAE	Day 21 #37	2003-11-22 17:36:56 [†]	332
TCAE	Day 21 #49	2003-11-03 11:51:44	265
CTCAE	No Fault	No Fault	No Fault
USAD	Day 21 #9	2003-11-01 11:11:44	264
ATF	Day 21 #37	2003-11-14 11:12:17	310
CTQN (CB)	Day 21 #1 [‡]	2003-11-01 01:21:44	455
AIRL ($\gamma = 0$)	Day 22 #163	2003-11-09 01:01:44	261
AIRL (Ours)	Day 22 #163	2003-11-17 12:22:30	459

GT: Provided Ground Truth. AIRL ($\gamma = 0$): Ablation results (Sec A). [†] Post-GT: Misses onset window under strict threshold. [‡] CTQN: Binary classification collapse (all-Faulty constant score ≈ 0.546).

onset. The ablated AIRL ($\gamma = 0$) flagged on Nov 9, behaving like the statistical baselines and confirming that the discount factor is what separates genuine degradation from operational noise.

On XJTU-SY (healthy window ends at ≈ 500 min [20]), every baseline except CTQN triggered between Files 264–332, more than 150 min before the onset boundary and well within the established healthy range; these are clear pre-onset false alarms. CTCAE failed to register any deviation. Two methods produced detections close to the onset boundary: CTQN at File 455 and AIRL at File 459. Given that the ~ 500 -min reference is itself an approximate onset annotation rather than a hard cutoff, both detections fall within a 45-min precursor window of the verified degradation transition, and are best interpreted as operationally reasonable early warnings rather than false alarms. On this dataset alone, CTQN’s bandit formulation produced a comparably reasonable detection point; however, its performance on HUMS (mode collapse) and IMS (false alarm Nov 1) shows this is not a stable property of the formulation. AIRL achieves comparable near-onset detection on XJTU-SY and stable behaviour on the other two datasets, which is the broader claim the bandit baseline cannot match.

5.2 Post-Detection Consistency

Beyond detection timing, a practical prognostic system must remain stable throughout the degradation period. *Post-detection consistency* measures the proportion of files correctly flagged as anomalous after the confirmed ground-truth onset [13, 15]; a high PDC indicates that the model tracks fault progression steadily without frequent dropouts. Critically, PDC must always be interpreted alongside detection timing: a method that triggers prematurely and never stops alarming will accumulate a near-perfect PDC through false-alarm persistence, not genuine discrimination. We term this *saturation failure*, and it renders the PDC score functionally meaningless for that method.

Table 4: Pre-GT FAR and PDC after Ground Truth (Otsu).

Method	HUMS		IMS		XJTU-SY	
	FAR%	PDC%	FAR%	PDC%	FAR%	PDC%
IF	[95.1]	[99.62]	57.2	59.70	50.3	50.52
OCSVM	[99.2]	[100.0]	60.3	61.86	58.9	59.18
AE	[100.0]	[100.0]	82.6	90.73	[97.8]	[98.55]
VAE	89.1	99.86	50.0	49.91	46.8	46.85
LSTM-AE	[93.2]	[99.24]	50.2	50.18	46.5	46.58
LSTM-VAE	[91.6]	[99.24]	50.2	50.12	46.5	46.58
TCAE	41.8	68.06	88.3	92.68	86.3	87.62
CTCAE	0.0	0.00	0.0	0.00	0.0	0.00
CTQN (CB)	[100.0]	[100.0]	52.7	90.0	63.0	100.0
USAD	[99.8]	[99.8]	70.1	70.9	81.4	81.5
ATF	[94.9]	[96.32]	11.6	94.3	[100.0]	[81.79]
AIRL (Ours)	42.2	79.09	88.1	90.15	68.4	68.38

[] denotes saturation failure (FAR > 90%); high PDC reflects false-alarm persistence, not genuine discrimination.

Two-threshold evaluation framework. Table 3 (earliest detection) uses the *max-2SE* threshold, which is calibrated from the maximum healthy training score and is one of the max-based strategies producing earliest within-window detection across all three datasets in Table 5; it answers “*when* does the method first alarm under strict, max-based calibration?” Table 4 PDC uses Otsu’s adaptive threshold, which partitions test scores globally to maximize between-class variance; it answers “*how reliably* does the method track fault progression after onset?”

HUMS2023 Gearbox: All reconstruction-based baselines achieve >99% PDC on HUMS under the Otsu threshold, but only because they trigger prematurely on Day 21 and flag virtually every subsequent file, collapsing to an “always-anomalous” state. USAD exhibits the same pattern across all three datasets. ATF saturates on HUMS (FAR=95%). CTCAE scores 0% PDC on HUMS, indicating total detection failure. CTQN scores 100% PDC on HUMS but with 100% FAR (all files flagged Faulty from Day 21 #1), confirming saturation failure rather than genuine detection; its high PDC is as meaningless as any other saturated method. TCAE achieves 68.06% PDC on HUMS, with its bearing-dataset numbers similarly inflated by pre-GT false-alarm persistence (Table 4).

IMS Bearings: On IMS, AIRL achieves PDC of 90.15% with a FAR of 88%. ATF achieves a low FAR (12%) on IMS but collapses to zero post-GT detections under strict thresholds, confirming its IMS score range is inadequately spread for conservative calibration. TCAE and IF achieve competitive PDC (92.68%, 59.70%) but at the cost of high pre-GT FAR (88%, 57% respectively), confirming that their IMS performance is inflated by persistent flagging begun well before degradation onset. CTQN achieves PDC of 90.0% with a FAR of 52.7% on IMS, with detection from Nov 1 (01:21:44), well before the mid-November onset, indicating its IMS PDC is partially inflated by early false-alarm persistence beginning from the first test day.

XJTU-SY Bearings: On XJTU-SY, ATF completely saturates (FAR=100%) while AIRL maintains a FAR of 68% with a PDC of 68.38%. The compressed score range of this bearing dataset makes Otsu partitioning less precise for all methods; nonetheless, AIRL

is the only method achieving consistent non-saturated behavior across this dataset as well. CTQN achieves PDC of 100% with FAR of 63.0% on XJTU-SY, with first detection at minute 455 (pre-GT), suggesting competitive post-fault tracking, but this is evaluated on a 16,295-file subset (the last 2,000 test files are withheld as training data), so the post-GT window differs from other methods.

AIRL (Ours) is the only method achieving meaningful, non-saturated PDC across all three datasets: 79.09% (HUMS), 90.15% (IMS), and 68.38% (XJTU-SY). On HUMS, AIRL significantly outperforms every non-saturated baseline. IMS is AIRL’s weakest dataset: the pre-GT FAR of 88% reflects that gradual bearing degradation produces a compressed AIRL score range, where Otsu’s adaptive split places the decision boundary close to the healthy distribution. ATF achieves a lower IMS FAR (12%) under Otsu but saturates on HUMS (FAR=94.9%) and XJTU-SY (FAR=100%), and collapses to zero post-GT detections under strict thresholds (Table 3); its IMS success is dataset-specific, not a general property. TCAE matches AIRL’s IMS FAR (88%) and also stays below the saturation ceiling, but its first detection is pre-onset on all three datasets (Day 21 #49, Nov 3, 265 min; Table 3), so its PDC reflects persistent early flagging. AIRL is the only method that combines within-window detection timing with non-saturated FAR on every dataset, demonstrating that the temporal reward signal generalises across distinct fault types where reconstruction and attention-based baselines do not.

Operational usability of these false-alarm rates: The FAR values in Table 4 are worst-case: they use Otsu, which we adopt only to make PDC comparable across methods and which Section 5.3 advises against for deployment. Under the recommended Max-2SE operating point the pre-onset FAR falls to 0.38% on HUMS (1 of 263 pre-onset files) and 0.10% on XJTU-SY (2 of 1,992), and to zero on IMS, where the max-based rules flag a single file in the whole test set and it lies inside the onset window. These are operationally usable rates, and they are unfiltered per-file counts: practical alarm policies add persistence rules, hysteresis, or score smoothing, which help only a non-saturated detector, since a method with FAR above 90% has excursions that are not isolated.

5.3 Threshold Sensitivity Analysis

Threshold choice is a practical deployment concern. Table 5 compares the earliest AIRL detection under all threshold strategy families described in Section 3.4.

On HUMS, five independent strategies converge to the same detection point (Day 22, #163): mean+3 σ , all three max-based variants, and GPD tail (p_{99}). Percentile p_{99} is also within the onset window (Day 22 #100). This cross-strategy consistency confirms that the AIRL reward signal produces a clean, threshold-invariant separation on gearbox data. Loose strategies (Otsu, k -means) trigger on Day 21 #2.

On IMS, the threshold landscape is sharply polarised. All loose and moderate strategies (mean+ $k\sigma$, median+3MAD, percentile, Otsu, k -means) trigger false alarms from Nov 1, well before the mid-November degradation window, while GPD tail (p_{99}) triggers on Nov 8. In contrast, all three max-based strategies (Max, Max-SE, Max-2SE) converge to exactly one anomalous file across the entire 8,940-file test set: Nov 17 12:22:30, precisely within the mid-November onset window. This binary behaviour, either saturation

Table 5: Earliest AIRL detection per threshold strategy

Threshold	HUMS (Day #)	IMS (date)	XJTU-SY (min)
Mean+2 σ	Day 21 #88 $\ddagger$	Nov 1 $\ddagger$	—
Mean+3 σ	Day 22 #163 $\dagger$	Nov 1 $\ddagger$	—
Median+3MAD	Day 21 #80 $\ddagger$	Nov 1 $\ddagger$	—
Max	Day 22 #163 $\dagger$	Nov 17 $\dagger$	—
Max-SE	Day 22 #163 $\dagger$	Nov 17 $\dagger$	533 $\dagger$
Max-2SE	Day 22 #163 $\dagger$	Nov 17 $\dagger$	459 $\dagger$
Percentile (p_{99})	Day 22 #100 $\dagger$	Nov 1 $\ddagger$	267 $\ddagger$
Otsu	Day 21 #2 $\S$	Nov 1 $\S$	251 $\S$
k -means midpoint	Day 21 #2 $\S$	Nov 1 $\S$	251 $\S$
GPD tail (p_{99})	Day 22 #163 $\dagger$	Nov 8 $\ddagger$	1748 $\dagger$
Ground truth onset	Day 24 #264	mid-Nov.	~500 min

$\dagger$ Valid: Within-window / post-GT. $\ddagger$ False Alarm: Pre-GT.

$\S$ Saturation: Flags first test file. —No Detection: Threshold not met.

or single-point detection, reflects the compressed AIRL score range on IMS, where the Nov 17 file is the sole test sample that marginally exceeds the healthy training maximum.

On XJTU-SY, the usable strategy space is narrower due to the compressed score range: mean+2 σ , mean+3 σ , median+3MAD, and Max yield no detection because no test file exceeds the corresponding threshold. The reported onset is approximately 500 minutes (end of the healthy window); the precise transition is subtle. Both Max-2SE (minute 459) and Max-SE (minute 533) detect within a 75-minute window of this boundary, with Max-2SE flagging the earliest pre-onset deviation and Max-SE confirming detection just after. For prognostic deployments where early warning matters, Max-2SE is preferable; for confirmed-fault detection, Max-SE is more conservative. GPD tail (p_{99}) is excessively delayed (minute 1748), and Otsu and k -means saturate from minute 251.

Across all three datasets, the max-based family (Max-SE and Max-2SE) provides the most reliable detection. Both variants produce identical detections on HUMS (Day 22 #163) and IMS (Nov 17), and bracket the XJTU-SY onset by under 75 minutes. We recommend Max-2SE as the default for prognostic early warning (earliest detection within the onset window across all three datasets), and Max-SE as the conservative alternative when post-onset confirmation is preferred. Both are self-calibrating from healthy training scores and require no held-out validation set or manual tuning.

5.4 Computational Cost and Deployment

Adversarial training is commonly cited as expensive; in practice, AIRL’s overhead is modest due to two structural properties. First, its three networks (reward, value, policy) contain only 644 parameters, one to four orders of magnitude fewer than the deep baselines (TCAE ~14K, LSTM-AE ~167K, ATF ~146K, AE ~9.0M, USAD ~13.4M). Second, at inference time AIRL operates on consecutive scalar pairs (x_t, x_{t+1}): each full-length file is reduced to a single transition pair before scoring, requiring no matrix multiplication over high-dimensional vectors. All reconstruction-based methods (AE, TCAE, USAD, ATF) must forward-pass the full sequence vector per file. This makes AIRL’s per-file inference constant, minimal, and compatible with edge controllers that cannot buffer complete sample windows. For deployment: (i) collect ≥ 250 healthy files;

(ii) train AIRL offline once on healthy transitions; (iii) at runtime, score each new acquisition file using the frozen reward and value networks; (iv) apply a Max-2SE threshold (or Max-SE for more conservative post-onset detection) calibrated on training scores for automated alarming.

6 Discussion and Conclusion

This paper presented a label-free, offline-trainable fault detection system based on adversarial inverse reinforcement learning, evaluated on three industrial run-to-failure benchmarks. While our primary objective remains achieving the *earliest possible detection*, AIRL was the only method to validate these early warnings by achieving non-saturated, consistent PDC (68–90%) across all three datasets while respecting onset ground-truth timelines. Reconstruction-based baselines (AE, VAE, LSTM variants, USAD) saturate indiscriminately; CTCAE collapses to zero detections on all three datasets; TCAE stays non-saturated but alarms pre-onset on all three, so its PDC reflects early flagging rather than timely detection; ATF saturates on HUMS and XJTU-SY, and collapses on IMS under strict thresholds despite low FAR under Otsu. These results confirm that higher reconstruction capacity does not prevent the saturation-or-collapse failure inherent to run-to-failure prognostics. AIRL’s temporal reward modeling is the key differentiator.

Scope of the RL claim: Our argument is precise and deliberate: we are not claiming reinforcement learning is universally superior to supervised or signal-processing methods for fault detection. Simple machine-learning and signal-processing baselines frequently match or outperform deep models in this domain [22, 37, 46], and supervised methods remain the right tool for fault *classification* once detection has flagged an anomaly [11, 51]. Our claim is the more specific one: *if* RL is to be applied to MFD, the prevailing contextual-bandit formulation defeats its purpose. Genuine RL, whether online or offline, is distinguished by reasoning over *state transitions* and credit-assignment through *delayed reward* [3]; an episode that terminates after a single observation reduces to supervised learning. To operationalize sequential reasoning without physical control actions, we adopted a state-only imitation learning perspective [40], framing the next observation as a proxy action. This recovers a reward function that measures the *plausibility of the transition itself*, shifting focus from “what does the signal look like?” to “how is the signal evolving?” The contextual-bandit baseline (CTQN), which lacks this temporal reasoning, exhibited mode collapse on HUMS (all files flagged Faulty, FAR=100%) and pre-GT false alarms on IMS, empirically confirming the theoretical argument; its near-onset detection on XJTU-SY (File 455) does not generalize.

Contribution: To our knowledge this is the first principled application of inverse RL to machinery fault detection, demonstrating that offline RL can be applied to industrial prognostic data without collapsing into supervised classification. The contribution is methodological: a transferable blueprint for using sequential RL on offline industrial sensor data, with an associated evaluation framework (timeliness, FAR, PDC, saturation flagging) that surfaces failure modes which standard metrics like AUROC hide.

Limitations and future work: Four limitations are worth naming explicitly. First, AIRL operates on single-channel scalar transitions $(x_t, x_{t+1}) \in \mathbb{R}^2$, which keeps the network compact and

edge-deployable but discards cross-channel correlations that multi-sensor industrial systems routinely provide; multi-sensor fusion with shared discriminator architectures is a logical extension. Second, AIRL’s performance is dataset-dependent: on IMS, the gradual nature of bearing degradation produces a compressed AIRL score range, which translates to a higher pre-GT FAR (88%) than on HUMS (42%) or XJTU-SY (68%). While AIRL remains the only method maintaining non-saturated FAR across all three datasets, the IMS result indicates that gradual bearing degradation is the harder setting for transition-based reward learning, and motivates non-stationary or distribution-shift-aware threshold adaptation. Third, validation is currently limited to three public run-to-failure datasets; larger industrial corpora, online deployment trials, and uncertainty quantification on the learned reward function remain future work. Fourth, the system detects but neither diagnoses nor acts. Because the setting is offline and purely observational, with no recorded control actions (Section 3.1), AIRL recovers a health reward from observed transitions instead of optimising a policy over maintenance actions. Consequently, it functions as a sequential anomaly detector and not a decision system. Closing that loop would require logged interventions and their outcomes, which no public run-to-failure benchmark currently provides.

A Ablation Study: Impact of Sequential Reasoning

To validate the role of sequential modeling, we ablated the discount factor (γ), comparing AIRL ($\gamma = 0.99$) against a myopic variant ($\gamma = 0$) that reduces the agent to a contextual bandit. The sequential formulation offers three measurable advantages, each targeting a distinct failure mode of the bandit formulation. (i) *Tighter healthy-manifold representation:* on HUMS2023, the sequential discriminator loss stabilized at ≈ 0.75 versus ≈ 0.83 for the myopic baseline, and the p_{99} anomaly threshold was lower ($\tau \approx 0.319$ vs. 0.347); despite the stricter threshold, the sequential model flagged a higher total anomaly volume, indicating that the value function characterizes healthy behavior precisely enough that subtle fault indicators register with high confidence rather than requiring large deviations. (ii) *Noise filtering via temporal credit:* on XJTU-SY, the myopic model triggered false alarms within the known healthy window (< 500 min), File 277 under max-SE and File 261 under max-2SE, whereas the sequential model filtered these fluctuations and delayed detection to File 533 (max-SE) and File 459 (max-2SE); without temporal credit assignment, the bandit formulation cannot distinguish transient instability from genuine degradation. (iii) *Stability across thresholds:* on IMS, the myopic baseline produced zero detections under strict thresholds (max, max-SE), and its earliest detection (Nov 9) required loose adaptive thresholds (GPD) and collapsed under stricter calibration, while the sequential model triggered on Nov 17, within the mid-November degradation window [4], and maintained dense, stable detection across the full threshold hierarchy. Their cumulative effect explains why AIRL’s full $\gamma = 0.99$ configuration consistently outperforms the myopic ablation across all three datasets reported in the main results.

Usage of Generative AI

The GenAI models (and Copilot) were used only to fix the LaTeX equations and formatting issues, refine sentences or words, fixing minor bugs in code. All the generated suggestions were thoroughly verified, edited, and rewritten by the authors, and no AI text, figures, or code appears in the manuscript. The authors maintain full responsibility and accountability for all concepts & methodology, data, analysis, and results & conclusions, and this limited use of GenAI complies with ACM's authorship and transparency policies.

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